

Build dependency graph — what parallelizes, what cannot, and why

Written 2026-08-07. Companion to docs/IMPLEMENTATION-PLAN.md, which says *what* to build and how long each step takes. This document says **what may run at the same time**, and it is written to be handed to an orchestrating agent.

The answer up front: the critical path is 7.75 h if task #28 is done cheaply, 15.75 h if it is done as originally specified, against ~36 h run as written today. Only 3.25 h of that is jobs on the path — code now dominates the schedule, which is a change from every earlier version of this document.

⚠ Two 2026-08-07 findings reshaped this graph. Both are MEASURED; neither was known when it was written.

- 1. The CPU cap is 384 vCPU, not 128 (IMPLEMENTATION-PLAN.md §8B.5 — quota 1,152, one queue 1:1, all 5 AZs offer the type). The tokenize floor is 2.21 h, not 6.61 h, and 12 c7i.8xlarge children fit rather than 4.
- 2. verify --deep is retirable (§8.3a) — a verified PUT is rejected server-side on digest mismatch, so the re-hash re-establishes a guarantee the write path already enforces. Needs B7 first (~5 lines).

And together they INVERT the C3b recommendation, which is the single most important thing on this page:

scenario	BUILD	code ahead of it	critical path
the graph as first written — C3b deferred, BUILD assumed 6.6 h	6.6 h	A2a 4 h	13.31 h ❌ never achievable
do nothing — DCLM stays one un-splittable child	10.85 h	A2a 4 h	16.39 h
C3b as specified — plan-time ordinal ranges, ~12 h of code	2.21 h	C3b 12 h	15.75 h — buys only 0.64 h
C3b via the domain_column fan-out, ~2 h of code	2.21 h	absorbed by A2a	7.75 h ✅

Break-even for C3b is ~11.5 h of code. As originally specified (12 h) it is within noise of not doing it at all; done cheaply it is worth 8.6 h. The instruction to stream 4 changes accordingly — see §6.

The full history of this number, because it moved three times and each move had a cause:

version	BUILD	path	what changed
as first written	6.6 h	13.31 h ❌	assumed a floor its own deferred item made unreachable
after blocker 0	6.6 h	21.31 h	C3b added at 12 h, on the path
after B7 / retiring VD1	6.6 h	20.14 h	–1.17 h (not –1.49: GA1 was parallel and gets exposed)
after the 384 vCPU measurement	2.21 h	7.75 h ✅ / 15.75 h	the floor fell 3×, which inverted how C3b should be done

Read the last row carefully: BUILD got 3× cheaper, so 12 h of code to protect it stopped being worth it. The 7.75 h figure requires the cheap route; the 15.75 h figure is the same graph with the expensive one.

The job-time floor is 3.25 h — SMOKE 0.4 + BUILD 2.21 + PUB1 0.3 + GA1 0.32 + PR1 0.02, the jobs actually on the path once VD1 is retired and BUILD runs at the measured 384 vCPU. (Earlier versions of this

line said 8.41 h, then 8.81 h, then 7.64 h, each summing a different set at a different cap; §5 derives it explicitly so it cannot drift again.)

Everything above 3.25 h on the critical path is code and the image build — which is why the C3b implementation choice, not any job flag, is now the largest single lever in this document.

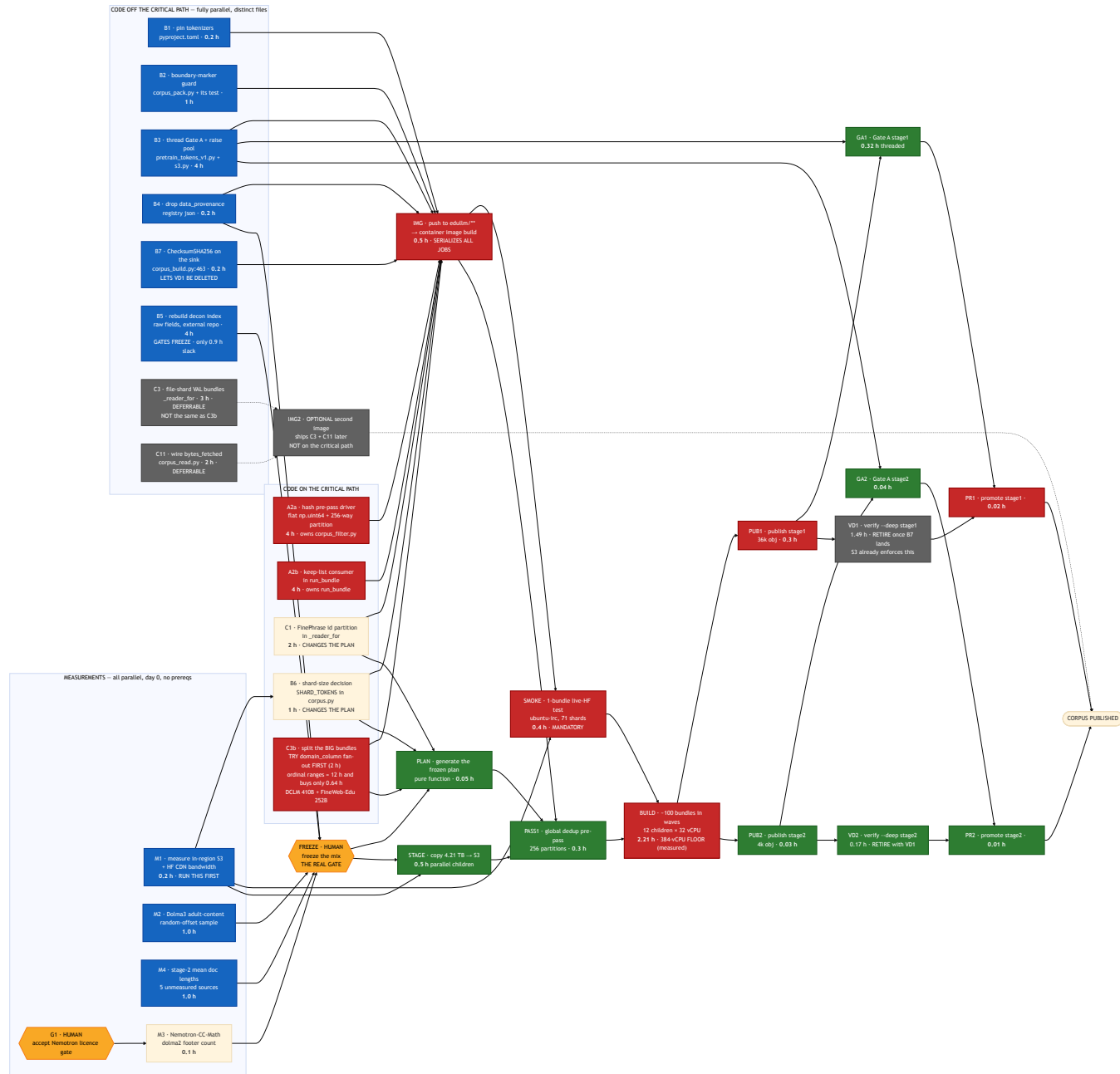
1. The three currencies, which are not interchangeable

Most parallelization mistakes here come from mixing these up.

currency	unit	what buys more of it	what caps it
agent-hours	code items	more agents in more worktrees	file contention (§3), not logic
wall-hours	AWS jobs	more array children, more workers	384 vCPU compute environment (MEASURED; it was never 128)
calendar	human gates	nothing — you wait	approval latency, license acceptance

A code item and a job of the same nominal duration are **not** substitutable: 8 agent-hours of code can run alongside 8 wall-hours of jobs, but two code items touching the same function cannot run alongside each other at all.

2. The graph



Red = critical path. Blue = parallel. Grey = deferrable to a second image. Amber = waits on a human.

3. ⚠️ The real constraint is file contention, not logic

This is the part an orchestrator gets wrong. Eleven code items look independent on a task list. They are not — five of them edit `corpus_build.py`, and within that file they cluster into **two functions**:

function	items that edit it	verdict
<code>_reader_for</code>	C1, C3, C11	one owner, or three-way conflict
<code>run_bundle</code>	A2b, C7	one owner
<code>allocate_ordinals</code> / <code>plan_document</code>	C3b, B6	one owner — and it is the critical-path item
<code>corpus.py</code>	B6, C3b, C3	3 — different constants and functions, same file

file	items	contention
<code>corpus_build.py</code>	C1, C3, C4→A2b, C7, C11	5 — hottest file in the plan
<code>corpus.py</code>	B6, C3b, C3	3 — and C3b is critical, so a conflict here costs image time
everything else	1 each	none

⚠️ **C3b and B6 belong to the same agent.** Both change what `plan_document` emits — B6 changes `SHARD_TOKENS`, C3b adds per-child ordinal ranges — and both therefore change every `plan_id`. Splitting them across two agents means two conflicting rewrites of the plan schema on the critical path. Give one agent `corpus.py`'s plan surface entirely.

Rules for the orchestrator:

1. **One agent owns one function, never one file.** `_reader_for` and `run_bundle` can be two agents in two worktrees, because they do not overlap textually — but both must be told the other exists.
2. **Distinct-file items are genuinely free.** B1, B2, B3, B4, B5 can be five simultaneous agents with zero coordination.
3. **Worktrees are mandatory** for anything touching `corpus_build.py`, per this repo's own convention:
`../Capstone_LLM-worktrees/edullm-data/<agent-id>--<task-slug>` on `agent/<agent-id>/<task-slug>`.
4. **Merge order is** `_reader_for` → `run_bundle` → **everything else**, because the first two are on the critical path and a late merge conflict there costs image-build time.

4. The four hard serialization points

Nothing parallelizes past these. They are why the path is 21.31 h and not the 8.81 h of job time inside it.

S1 — the image build gates every AWS job

Container images build **only from** `edullm/**` branches (`.github/workflows/edullm-platform-build.yml:9-10`). A merge to `main` builds **nothing, silently**, and a submission naming that commit is refused for having no image. So **all code that the jobs need must land in one push**, and every job waits on it.

Consequence: batching code into one image is *better* than shipping two. Measured against the graph, a two-image scheme (pre-pass early, build later) came out **0.1 h worse** — the second build's 0.5 h lands directly on the critical path.

S2 — `FREEZE` is a human decision with an unbounded duration

Adding a source after the plan is generated renames **98% of shards** and voids **882B tokens** (`IMPLEMENTATION-PLAN.md §0`). So the mix must be final first. **Its duration is a decision, not a computation** — it is the one node whose length no amount of parallelism touches.

S3 — `BUILD` is capped at 384 vCPU (measured; it was never 128)

2.21 h is the tokenize floor at the real cap: $1.0T \div (384 \times 0.328 \text{ M tok/s/vCPU})$. Verified 2026-08-07 — `maxvCpus: 384`, EC2 quota 1,152, one queue 1:1, `c7i.8xlarge` offered in all 5 of the CE's AZs (`IMPLEMENTATION-PLAN.md §8B.5`). The cap allows **12 concurrent children**, not 4.

⚠ **But the floor is only reachable if the big bundles are split, and more capacity makes that WORSE.** Per-child duration is *that child's* tokens $\div$ *that child's* vCPU, and a Batch child cannot exceed one instance. **DCLM is 410B in one non-fanning-out bundle — 10.85 h on its 32 vCPU, unchanged by the cap.** Against a 2.21 h aggregate floor that child is now **4.9× the floor**, where at 128 vCPU it was 1.6×.

FineWeb-Edu is the second one, and it is easy to forget: 252B = **6.67 h** on one instance. **Splitting DCLM alone leaves FineWeb-Edu binding at 6.67 h and the path at 12.21 h**, not 7.75 h. Whichever route stream 4 takes must cover both.

****And `--shard/--of` strides *bundles*, so the capability does not exist in the code**** — see `IMPLEMENTATION-PLAN.md §8A.5a`, including the cheap `domain_column` alternative that now decides whether this whole item is worth 0.64 h or 8.64 h.

S4 — `~~ VD1 cannot start until PUB1 finishes~~` → **delete VD1 instead**

`verify --deep` re-hashes published objects, so it is strictly after publish. At 8 workers it is 1.49 h and it sits on the critical path; at `--hash-workers 1` it is 11.7 h and exceeds its timeout.

⚠ **But the right move is to retire it, not thread it** (`IMPLEMENTATION-PLAN.md §8.3a`). A verified PUT is rejected server-side on digest mismatch — proven live 2026-08-01: `BadDigest`, and **no object is created** — and `CopyObject` recomputes the checksum on the publish hop. So the re-hash re-establishes a guarantee the write path already enforces.

Prerequisite: `B7.corpus_build.py:463`'s sink calls a plain `put` and declares no checksum, so today the re-hash is the only thing behind those bytes. **Ship B7 (~5 lines) first, then delete VD1**. Never the reverse.

Effect: critical path 21.31 h → 20.14 h. Note that is **1.17 h, not the 1.49 h VD1 costs** — `VD1` and `GA1` run in parallel between `PUB1` and `PR1`, so deleting the longer branch **exposes** the shorter one. Deleting a parallel branch buys the *difference*, not the branch. Recomputed on the DAG, not subtracted.

S4 therefore stops being a hard serialization point at all — the remaining PUB1 → GA1 → PR1 chain is 0.34 h.

5. The critical path, and everything with slack

Critical path — 7.75 h with the cheap C3b route and VD1 retired. The expensive C3b route gives 15.75 h; the difference is 12 h of code minus 2 h, sitting in front of IMG.

from → to	node	why it cannot move
0.00 → 4.00	A2a hash pre-pass driver	the longest <i>unavoidable</i> code item; C3b's cheap route hides inside this 4 h
4.00 → 4.50	IMG image build	S1 — one push, edu11m/** only
4.50 → 4.90	SMOKE live-HF smoke test	mandatory; the path has never run
4.90 → 7.11	BUILD ~100 bundles	S3, the 384 vCPU floor — 2.21 h
7.11 → 7.41	PUB1 publish stage 1	after build
7.41 → 7.73	GA1 Gate A stage 1	recomputes from bytes in a different process — the one gate that is not a tautology (IMPLEMENTATION-PLAN.md §8.3b)
7.73 → 7.75	PR1 promote stage 1	after the gate

With the expensive C3b route, C3b's 12 h replaces A2a's 4 h at the head and everything shifts +8 h. That single substitution is the whole 7.75 h ↔ 15.75 h spread.

Where the 3.25 h job floor comes from, stated as a sum so it cannot drift from the headline again:

node	h	on the path?
SMOKE	0.40	✓
BUILD	2.21	✓ — at 384 vCPU, was 6.60 at the withdrawn 128
PUB1	0.30	✓
GA1	0.32	✓ — it inherits VD1's slot on the path
PR1	0.02	✓
subtotal — the job floor	3.25	
STAGE 0.5 · PASS1 0.3 · PLAN 0.05	0.85	✗ absorbed by the code prefix's slack
PUB2/GA2/PR2	0.08	✗ parallel to stage 1's tail
~~VD1 1.49 · VD2 0.17~~	~~1.66~~	retired — S4
all remaining job rows summed	4.18	—

⚠️ GA1 moved onto the critical path by deletion, not by getting slower. It always ran in parallel with VD1 between PUB1 and PR1 ; removing the 1.49 h branch exposes the 0.32 h one. That is why the saving is 1.17 h and not 1.49 h.

And it makes B3 (thread Gate A) a precondition for B7 paying off at all. Computed on the DAG:

	GA1 threaded (B3 done)	GA1 unthreaded (5.08 h)
keep VD1	21.31 h	24.90 h
delete VD1	20.14 h ✓	24.90 h

Without B3 , GA1 's 5.08 h dominates and deleting VD1 buys **exactly nothing** — the path is 24.90 h either way. So B7 is not harmful on its own, it is simply **inert until B3 lands**. Ship B3 first, or ship them together; sequencing B7 ahead of B3 produces a change with no measurable effect and invites the conclusion that the analysis was wrong.

The remaining 12.5 h of the path is C3b (12.0) + IMG (0.5) — code and image, not jobs. An earlier version of this document quoted an 8.41 h floor that matched neither the job subtotal nor the full sum; this table replaces it.

⚠️ A2a is no longer the path's head. At 4 h it now finishes inside C3b's 12 h, so the pre-pass has 8 h of slack — but it must still land before IMG , so it is not deferrable, merely no longer critical.

Everything else has slack and should be started at t=0 regardless:

node	duration	slack	note
M1 bandwidth	0.2 h	large	but run it first anyway — it calibrates every other estimate
M2 / M4 samples	1.0 h each	~11 h	gate FREEZE , not the build
M3 footer count	0.1 h	~11.9 h	blocked on a human licence acceptance (G1)
A2a hash pre-pass driver	4 h	8 h	was the path's head at 13.31 h; C3b displaced it
A2b keep-list consumer	4 h	8 h	must finish before IMG, same as A2a
B3 thread Gate A	4 h	~15.5 h	only GA1 / GA2 need it
B5 rebuild decon index	4 h	~8.9 h	gates FREEZE . Was ~0.9 h at the old critical path — start it early anyway, the slack is a by-product of C3b being long, not of B5 being cheap
B1 / B2 / B4	≤1 h	large	trivially parallel
C3 file-shard VAL bundles	3 h	∞ — deferrable	pure read-volume saving. Not the same item as C3b, which is critical
C11 wire bytes_fetched	2 h	∞ — deferrable	instrumentation. Do not put it in the first image

Stage 2 is entirely parallel to stage 1's tail. PUB2 / GA2 / VD2 / PR2 total 0.25 h and finish while stage 1 is still verifying, so stage 2 contributes **nothing** to the critical path.

6. What the orchestrator should actually launch

Wave 0 — immediately, 8 concurrent workstreams

stream	work	kind
1	M1 bandwidth measurement	job — do this first, it recalibrates the rest
2	M2 Dolma3 sample + M4 doc lengths	job
3	~~Ask the owner to accept the Nemotron licence gate~~  DONE — measured at 134.0B by a teammate with access	—
4	C3b + B6 — split the big bundles and the shard-size constant. Owns <code>corpus.py</code> 's plan surface : <code>allocate_ordinals</code> , <code>plan_document</code> , <code>SHARD_TOKENS</code> . Try the cheap route first — it is worth 8.64 h; the expensive one is worth 0.64 h	agent, worktree
5	A2a hash pre-pass driver (owns <code>corpus_filter.py</code>)	agent, worktree
6	A2b keep-list consumer (owns <code>run_bundle</code>)	agent, worktree
7	C1 FinePhrase id partition (owns <code>_reader_for</code>) + B5 rebuild the decon index (external repo)	agent, worktree
8	B1 + B2 + B3 + B4 (four distinct files)	agent(s)

 **Stream 4 is new and it is the longest item.** An earlier version of this table had no node for it and assumed `BUILD` = 6.6 h anyway. It cannot be split from B6 — both rewrite what `plan_document` emits.

Do not launch C3 or C11 in wave 0. They are deferrable, they contend with C1 on `_reader_for`, and including them adds ~5 h of agent time to the critical path for zero wall-clock benefit. **Note C3 (val bundles) and C3b (big bundles) are different items** — the first is deferrable, the second is critical.

Wave 1 — merge, then one push

Merge in order `_reader_for` → `run_bundle` → the rest. Push to an `edullm/**` branch. **One image.**

Wave 2 — the gate

`FREEZE` the mix, generate the plan, stage the sources. `STAGE` can overlap `SMOKE`.

Wave 3 — jobs

`PASS1` → `BUILD` (16 children × 8 vCPU) → then stage 1 and stage 2 publish/validate **in parallel**.

Every AWS job needs a separate human release. Nothing auto-publishes.

7. Where parallelism is wasted — spend nothing here

tempting	why it does not help
More than 12 build children at 32 vCPU	384 vCPU ÷ 32 = 12. More children just queue. (The old "16 × 8 vCPU" shape was sized on the withdrawn 128 cap and gave the big bundles too little vCPU each)
Splitting <code>BUILD</code> below 2.21 h	that is the CPU floor at the measured cap, not a scheduling artifact
Threading <code>verify --deep</code> instead of deleting it	S4 — S3 already enforces the guarantee on a verified PUT. Optimizing it is work spent keeping a redundant job
Raising <code>maxvCpus</code> past 384	possible (1,152 of quota, 1,060 free) but pointless until the big bundles split — the un-splittable child is the constraint, not the cap
Two container images	measured 0.1 h worse — the second build lands on the critical path
Parallelizing <code>STAGE</code> harder	it has ~4 h of slack; it is never the constraint
Doing C3 / C11 now	deferrable, and they contend on the hottest function in the repo
More agents on <code>corpus_build.py</code>	file contention. A third agent there produces conflicts, not speed
Rushing before M1 lands	every read estimate is calibrated on an unmeasured bandwidth

8. The orchestrator brief — copy this

Hand this to the agent that runs wave 0. It encodes the constraints above as rules rather than prose.

You are orchestrating the Phase-0 build for `pretrain/final-dataset`. Launch these 8 streams concurrently and hold them to these rules.

Stream 1 (job, FIRST): measure in-region S3 read bandwidth and HF CDN throughput. Report both. Every read estimate in `IMPLEMENTATION-PLAN.md §8A` is calibrated on an unmeasured ~ 85 MB/s borrowed from an S3 measurement; one plausible reconciliation says the CDN is ~ 8.4 MB/s. If it is, tell me before anything else proceeds — it changes the staging decision.

Stream 2 (job): Dolma3 adult-content prevalence at **random offsets** (a prior attempt could not separate signal from HuggingFace preview ordering), plus mean document length for the 5 unmeasured stage-2 sources.

Stream 3 — CLOSED. The Nemotron-CC-Math licence gate is accepted and the count is **MEASURED** at **134.0B** under dolma2 ($3 \approx 83.6B + 4plus \approx 50.4B$). **Two things remain: record the exact `text_column` and `id` column names in writing before the registry row is written** (§4.2 is the cautionary case — two plausible `text` columns, wrong one picked silently), and **keep `4plus_MIND` out of the pool** — it is a rewrite of `4plus`, so including both double-counts.

Streams 4–7 (code, one worktree each, ONE FUNCTION each — NOT one file each): - 4 — the largest single lever in the plan, and it is a **CHOICE**, not a task: split the big bundles and settle `SHARD_TOKENS`. Owns `corpus.py`'s **plan surface** — `allocate_ordinals`, `plan_document`, `SHARD_TOKENS`. Both change every `plan_id`, so they cannot be two agents.

Why it exists: `--shard/--of` strides *bundles*, so DCLM's 410B is one child at 10.85 h even on a whole 32-vCPU instance — and a Batch child cannot exceed one instance, so no vCPU allocation fixes it. FineWeb-Edu's 252B is 6.67 h and is the second one; cover both or the floor is not reached.

⚠ Read `IMPLEMENTATION-PLAN.md §8A.3` and `§8A.5a` BEFORE writing code. The route decides everything:

route	code	critical path	--- --- ---	synthetic <code>domain_column</code> fan-out — DCLM's parquet mirror is 27,938 files; a file-index-derived <code>domain</code> makes <code>allocate_ordinals</code> emit separate bundles with no new mechanism	~ 2 h	7.75 h ✓	plan-time ordinal ranges, as originally specified	~ 12 h	15.75 h	do nothing	0	16.39 h
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The expensive route buys 0.64 h over doing nothing. The cheap one buys 8.64 h. Break-even is ~ 11.5 h of code. **Try the cheap route first and time-box it.** Its cost is a schema smell — a `domain` label that is not semantically a domain, inside `manifest_sha256` and unbackfillable — which is a real objection worth raising with the owner, not a reason to silently pick the 12 h route. **If the fallback looks like more than ~ 11 h, stop and say so:** at that point doing nothing is competitive, and that is a finding rather than a failure. - 5: the flat-`np.uint64` hash pre-pass. Owns `corpus_filter.py`. **Size it for DCLM at 325M documents / 27.92 GB as a set (§5.2a), not for `finephrase-table`** — the plan's own table omitted the worst bundle. - 6: the keep-list consumer. Owns `run_bundle` in `corpus_build.py`. - 7: the FinePhrase id partition. Owns `_reader_for` in `corpus_build.py`. **Ship it together with the reader-budget division by the keep fraction** — separately, every bundle finishes and then fails `verify` on unfilled refs. **And do not "fix" `_CHARS_PER_TOKEN` while you are in there; that change is withdrawn and points the opposite way (§3.1).**

⚠ **Streams 6 and 7 both edit `corpus_build.py` in different functions.** Tell each about the other. Worktree convention: `../Capstone_LLM-worktrees/edullm-data/<agent-id>--<task-slug>` on `agent/<agent-id>/<task-slug>`. Merge order is **stream 4 → stream 7 → stream 6 → the rest.**

Also stream 7 (code, external repo, no conflict): rebuild the decontamination index from `raw` benchmark fields (question alone, question + each choice, question + correct answer) **in addition to** the rendered form. **It gates the mix freeze — do not let it start late.**

Stream 8 (code, five distinct files, no coordination needed): pin `tokenizers` in `pyproject.toml`; fix the boundary-marker prefix guard in `corpus_pack.py` **and** the test that asserts the table length is 1; thread the Gate A profile checks in `pretrain_tokens_v1.py` **and** raise `max_pool_connections` in `s3.py` (threading against a 10-connection pool self-throttles); drop `data_provenance_initiative` from the registry; **and B7 — declare `ChecksumSHA256` on the shard upload at `corpus_build.py:463`, which today calls a plain `put` and computes the digest one line too late.**

⚠ **B3 (thread Gate A) and B7 are a pair, in that order.** B7 is what lets VD1 be deleted, but deleting VD1 exposes GA1 — and at GA1's unthreaded 5.08 h the path is 24.90 h whether VD1 is there or not, so **B7 is inert until B3 lands.** Ship B3 first or ship both together. Do not report B7 as a win on its own; it will measure as nothing.

Do NOT start: val-bundle file-sharding or the `bytes_fetched` wiring. Both are deferrable, both contend with stream 6 on `_reader_for`, and including them adds ~5 agent-hours to the critical path for **zero** wall-clock gain.

When all streams land: merge in the stated order, push **once** to an `edullm/**` branch — images build only from that namespace, a merge to `main` builds nothing silently — and stop. **Do not submit any AWS job.** Report what landed and wait for a release decision.

Standing rules: write findings to disk incrementally (a prior wave lost 4 agents to an API budget cap and only on-disk partials survived); grade every number MEASURED / DERIVED / CARD / UNVERIFIED; "*nobody has measured this*" is a finding, not a failure; and if you find that one of my numbers is wrong, say so — two of them already were.

9. Honest limits of this graph

- **Durations for code items are estimates, not measurements.** The *ordering* is well-evidenced; the absolute agent-hours are my judgement. An earlier session's per-bundle ETAs were all retracted by their own author with the verdict "*every per-bundle ETA I gave was wrong, in both directions.*"
- **BUILD's 2.21 h assumes linear vCPU scaling**, measured only at 32 vCPU. Never tested at 384 — and the extrapolation is now 12× rather than 4×, so this caveat carries more weight than it did.
- **12 concurrent `c7i.8xlarge` has never been demonstrated.** `desiredvCpus` is 0; the quota (1,152) and AZ coverage (all 5) support it, but obtainability is UNVERIFIED. Request the full wave shape once in `SMOKE`.
- **The 1800 s auto-cancel rule is fail-open for capacity failures** — Batch leaves `statusReason` null, so the rule never matches and a starved job sits in `RUNNABLE indefinitely, with no error` (`IMPLEMENTATION-PLAN.md §8B.7`). Set your own wall-clock expectation per job; the queue will not fail fast.
- **A non-Batch "lane" path draws on the same EC2 quota and is invisible to `batch list-jobs`**, so "the queue is empty" is not "the quota is free."
- **Every read duration assumes ~85 MB/s borrowed from an S3 measurement.** If the HF CDN is really ~8.4 MB/s, `STAGE` moves from 0.5 h to days and becomes critical. **M1 exists to settle exactly this — which is why it is wave 0, stream 1.**

- **The 13-gram decontamination scan's CPU cost has never been measured** (~193 billion Python-level `blake2b` calls). If it is comparable to tokenization, `BUILD` is longer than 6.6 h.
- `FREEZE` and `G1` **have no duration here because they are human**. In practice they may dominate the calendar, and no amount of parallelism touches them.